Assignment 2: Integer Programming

MGSC 662: Decision Analytics

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Instructions

Remark: Assignment 2 has two parts which you need to model and implement in Python using *Gurobi* package. The data is provided in a Jupyter Notebook file "Assignment 2-Work Sheet.ipynb". Please report your results in your Notebook file with codes in it, and clearly outline your answers. Lack of clarity may affect your grade and complaints on this matter will not be entertained.

1 The Apportionment Problem: It's Political

The problem of fairly allocating parliamentary seats is a political matter in democratic societies[1]. Suppose a country with a total population of $N = \sum_i p_i$ tries to allocate its s available seats to its provinces, each of which has a population of p_i. The "fair" way to distribute these seats is to allocate each province a seat quota proportional to its population:

$$q_i = s \times \frac{p_i}{N} = s \times \frac{p_i}{\sum_i p_i}.$$

Although this solution seems appealing, it ignores the fact that the number of allocated seats should be an **integer**. For instance, in a country with 30 million population and 130 parliament seats, seat quota for a province with a population of 5 million is $q_i = \frac{5 \times 130}{30} = 21.\overline{66}$, which is not possible. The question then is how to allocate the seats "fairly". There are two methods to address this question. The first is a widely-used method that minimizes the sum of the gaps between actual and ideal allocations. The second one minimizes the maximum gap.

Dataset

The parliament of a fictitious country has 601 seats to be allocated to 20 provinces with different populations.

The dataset is in Table 1 (it is also available in the Jupyter Notebook for this assignment.) The quota variable (q_i) is rounded to 5 decimals and available in vector "Quota" in the Notebook. (To make sure everyone is working with the same data, do not calculate the quotas again yourself.) Also, to prevent indexing issues when coding, we indexed provinces from $P0$ to $P19$.

Questions

1.1 Minimize the sum of absolute gaps (15 pts).

In this method, the goal is to minimize the sum of absolute gaps between the number of allocated seats to each province and its quota while leaving no seats unfilled. That is, we seek to find

[1] If you are interested, you can find more info on this Wikipedia entry.

Table 1: Dataset

Province	Population	q_i
P0	5300000	21.80219
P1	9200000	37.84531
P2	5400000	22.21355
P3	8000000	32.90897
P4	3200000	13.16359
P5	8800000	36.19986
P6	8100000	33.32033
P7	10100000	41.54757
P8	3600000	14.80903
P9	6200000	25.50445
P10	11700000	48.12936
P11	5800000	23.85900
P12	7000000	28.79535
P13	6300000	25.91581
P14	4800000	19.74538
P15	10700000	44.01574
P16	7000000	28.79535
P17	9400000	38.66804
P18	6400000	26.32717
P19	9100000	37.43395
Total	146100000	601

allocations a_0 to a_{19} to

$$\begin{aligned} \min_{a_0,\cdots,a_{19}} \quad & \sum_{i=0}^{19} |q_i - a_i| \\ \text{s.t.} \quad & \sum_{i=0}^{19} a_i = s. \end{aligned} \tag{1}$$

This is a constrained nonlinear program. But one can "linearize" this program which makes it possible to use a mixed integer programming algorithm to estimate the optimal variables. To do so, define $z_i = |q_i - a_i|$ for $i = 0, \cdots 19$. So now we can rewrite the original program in (1), as

$$\begin{aligned} \min \quad & \sum_{i=0}^{19} z_i \\ \text{s.t.} \quad & z_i = |q_i - a_i| \qquad i = 0, \cdots 19 \\ & \sum_{i=0}^{19} a_i = s. \end{aligned}$$

Although the objective function is now linear, we have nonlinear constraints. To linearize the constraints, we can substitute $z_i = |q_i - a_i|$ with two constraints: $z_i \geq q_i - a_i$ AND $z_i \geq -(q_i - a_i)$. Recall that a_i's (allocation variables) are integer variables, and z_i's (gap variables) are continuous variables.

(a) Explain why this linearization is valid. How many decision variables and how many constraints does this mixed integer program have?

(b) Find the minimum sum of absolute gaps using the optimal allocation. Your code must be clear and easy to parse.

1.2 Minimax (15 pts).

A second approach to this problem is that instead of the sum of absolute gaps, one can minimize the absolute value of the largest gap while leaving no seats unfilled. In other words, find the

allocations a_0 to a_{19} to

$$\begin{aligned} \min_{a_0,\cdots,a_{19}} \quad & \max_i |q_i - a_i| \\ \text{s.t.} \quad & \sum_{i=0}^{19} a_i = s. \end{aligned} \tag{2}$$

This program is not linear either. To linearize it, define $z = \max_i |q_i - a_i|$. So the original program turns into:

$$\begin{aligned} \min \quad & z \\ \text{s.t.} \quad & z = \max_i |q_i - a_i| \\ & \sum_{i=0}^{19} a_i = s. \end{aligned}$$

The objective function is linear but we now have nonlinear constraints. We can linearize these constraints by substituting them with $z \geq q_i - a_i$ AND $z \geq -(q_i - a_i)$ for $i = 0, \cdots 19$. This turns the program into the following mixed integer program:

$$\begin{aligned} \min \quad & z \\ \text{s.t.} \quad & z \geq q_i - a_i \qquad i = 0, \cdots 19 \\ & z \geq -(q_i - a_i) \qquad i = 0, \cdots 19 \\ & \sum_{i=0}^{19} a_i = s. \end{aligned}$$

Notice that a_i's (allocation variables) are integer variables, and z (gap variable) is a continuous variable.

(a) Explain why this linearization is valid. How many decision variables and how many constraints does this mixed integer program have?

(b) Find the maximum absolute gap between an allocation and quota in your optimal solution. Your code must be clear and easy to parse.

1.3 Political considerations (15 pts).

Due to political reasons, some considerations should be addressed when allocating seats to each province. Here, we want to implement three of them:

(i) The number of seats allocated to the two least populated provinces ($P4$ and $P8$) must be equal.

(ii) The number of seats allocated to each of the provinces with more than 10,000,000 population ($P7, P10,$ and $P15$) should not exceed its quota.

(iii) If the number of seats allocated to $P0$ is more than its quota, then the number of seats allocated to $P2$ must be more than its quota as well.

Based on the first method (minimize the sum of absolute gaps while filling all the seats) and by implementing these considerations as new constraints, find the answers to the following questions.

(a) Write down the mathematical formulas related to each consideration. How many decision variables and how many constraints does this mixed integer program have in total?

(b) Find the minimum sum of absolute gaps using the optimal allocation. How much is the difference between your answer to this part and your answer to 1.1-(b)? Your code must be clear and easy to parse.

1.4 Opinions (2 pts).

Why is that in the US democracy, California with a population of 39.52 million and South Dakota with a population of 900 thousand are each represented by two senators in the US Senate?

2 Did You Miss BalancedMilk Co. and Ms. Fraser?

The next two problems are based on Assignment 1 and Quiz 1 of this course.

2.1 The salesforce problem (15 pts).

Do you remember the firm managed by Ms. Fraser and her efforts toward finding the best way to distribute the $10,000 award among its salespeople[2]? Another option that Ms. Fraser and Mr. Lei want to evaluate has the following properties:

1. They want to maximize the average payoff.
2. The true matrix is the second one (**B**).
3. Each salesperson should receive at least 1500 units of payoff.
4. This limit can be dropped to 1000 units of payoff **for at most three salespersons**.
5. **At least three salespeople** should receive 2500 or more units of payoff.

Using the MIP methods, see if this policy is feasible or not. If it is feasible, what is the average payoff, and what is the payoff of each salesperson?

2.2 BalancedMilk Co., in the middle of the crisis (20 pts).

We have been through the problems the firm faces after it lost its main supplier, S_8[3]. The issue is not resolved yet, and S_8 is still shut down. But now BalancedMilk hired a new analyzer who is a student at McGill's MMA program and is capable of implementing the following constraints while minimizing the total cost:

1. The firm is **not** committed to distributing the milk supplied by the remaining supply centers, S_1 to S_7. (It has the option not to distribute the milk it receives from the remaining supply centers.)
2. The cost of not supplying milk to each demand center is 15$/tonne.
3. S_3, S_4, and S_5 belong to the same company. The company has imposed the following constraint: BalancedMilk should utilize $\geq 50\%$ (equal or more than 50%) of the capacity of at least two of the three supply centers owned by this company.
4. Because D_5 and D_6 are close to each other, the firm has some flexibility in answering their demands. So either D_5 receives $\geq 80\%$ of its demand, or D_6 receives $\geq 60\%$ of its demand, or both.
5. A competitor is trying to damage BalancedMilk's reputation by focusing on the discrimination it imposes between different demand centers. This competitor is trying to take advantage of the difference between D_2 and D_8 and D_9. Therefore, if D_2 receives $> 50\%$ of its demand, then D_8 and D_9 should both receive $\geq 50\%$ of their demand as well.

Now, as a capable MMA student at McGill, see if an optimal solution with these constraints exists. If yes, what are the lowest cost, the utilized capacity of each supply center, and the satisfied demand of each demand center?

2.3 BalancedMilk Co., before the next crisis (20 pts).

The issue with S_8 has been resolved and now the balance between supply and demand in the supply chain is established. But the managerial team at BalancedMilk is worried about possible future incidents. They are realistic enough to know that this was not the last time an outbreak challenges their supply chain. Therefore, they want to devise a **contingency plan** to be prepared for possible future crises.

[2] You may review Quiz 1 for the details of this problem
[3] You may review Assignment 1 for the details of this problem

At the very first step, they are interested in the best and worst possible cases that may happen and the cost associated with each of them. Once again, they put their trust in a capable MMA student from McGill to calculate the minimum and maximum cost of a contingent crisis. They provide the following list of information:

1. The firm **is** committed to distributing all the milk supplied by the active supply centers. (It must distribute all the milk it receives from the remaining supply centers.)

2. In the best case scenario, just one supply center will be shut down. In the worst case scenario, three supply centers will be.

3. The cost of not supplying milk to a center is 30$/tonne.

Which supply center is shut down in the best case? Which three supply centers are shut down in the worst case? What is the cost in each case?

N.B.: In the best case scenario, you are simply looking for a supply center the shut down of which results in the lowest increase in the optimal cost. But in the worst case scenario, the firm is looking for a set of three supply centers to shut down the minimal cost of which is higher than the minimal cost for any other set of three supply centers. Suppose, for instance, the firm wants to see if losing $\{S_1, S_2, S_3\}$ is worse than losing $\{S_4, S_5, S_6\}$. It will calculate the optimal allocation (the allocation with the minimal cost) in both cases, and then report the one with the higher minimal cost as the worse case. Therefore, you are dealing with a Minimax problem. Maybe you need to consider all possible combinations. Or maybe there is a better way to do this. Who knows?